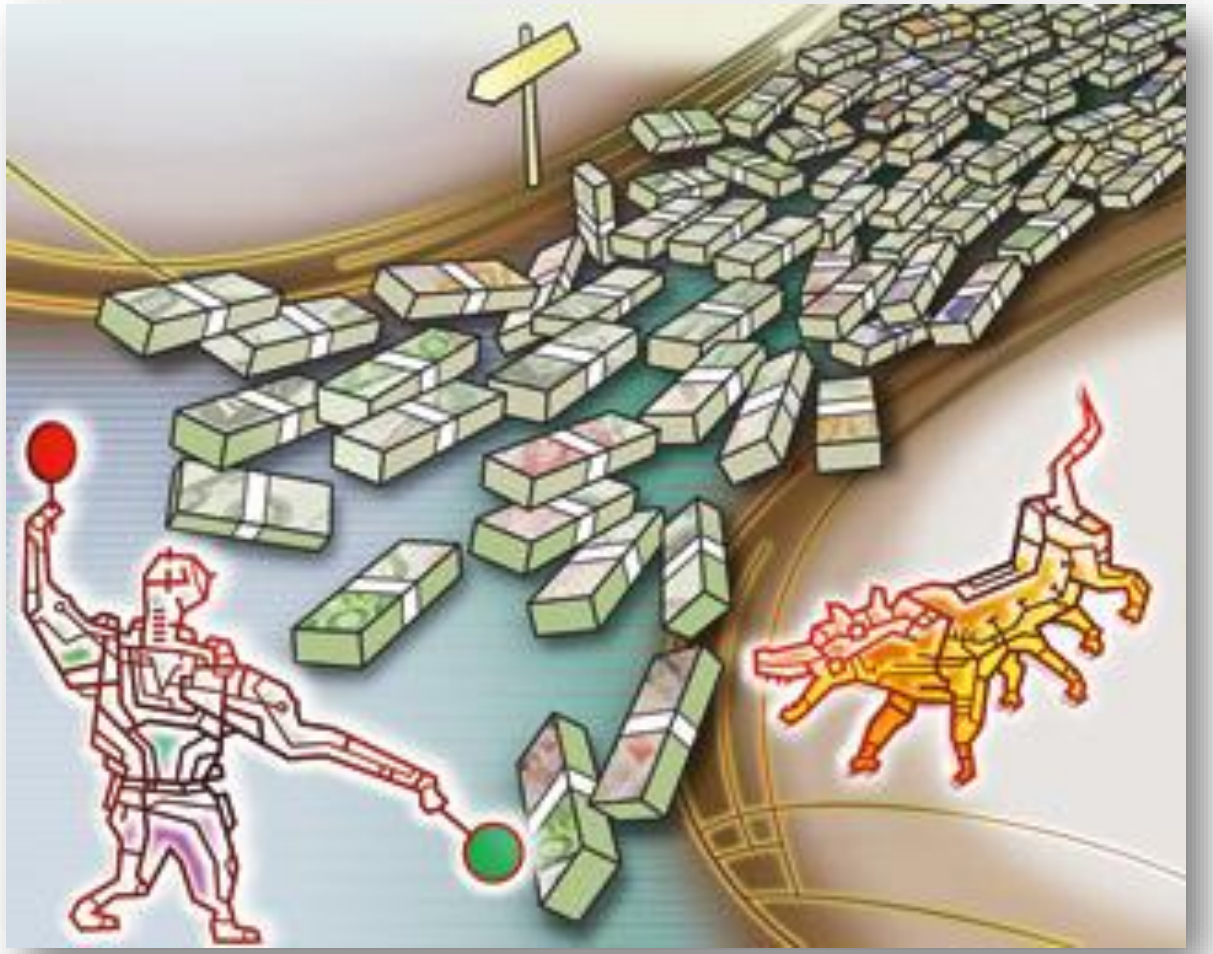


Lecture: International Finance (under fixed exchange rate)

Recap from last week:

Running persistent current/capital account deficit or surplus can lead to problems.

Some theories suggest that flexible exchange rate can ensure current/capital balance in the long-term

Recap from last week:

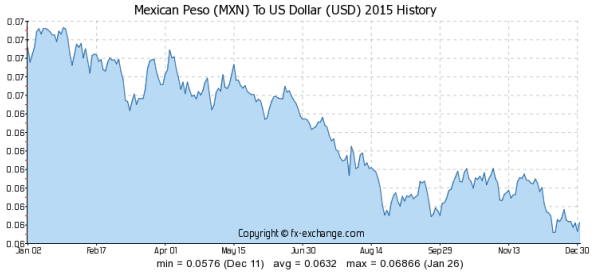
Running persistent current/capital account deficit or surplus can lead to problems.

Some theories suggest that flexible exchange rate can ensure current/capital balance in the long-term

But flexible exchange rates create instability for trade and investment.

<u>Supply of dollars</u>
Demand for pesos

<u>Demand for dollars</u>
Supply of pesos



ice INTERNATIONAL CURRENCY EXCHANGE

Recap from last week:

Running persistent current/capital account deficit or surplus can lead to problems.

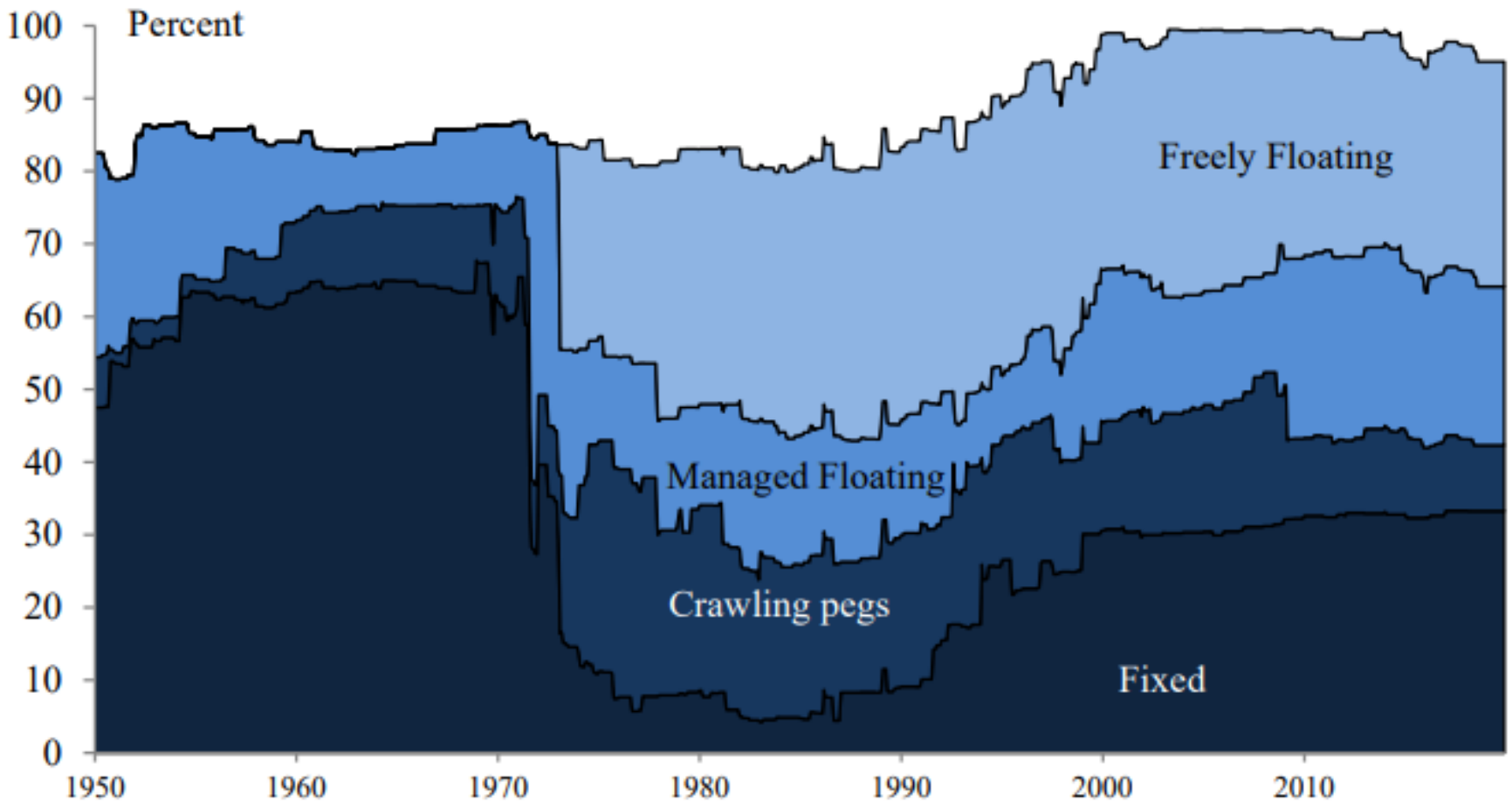
Some theories suggest that flexible exchange rate can ensure current/capital balance in the long-term

But flexible exchange rates create instability for trade and investment.

Fixed exchange rates provide you that stability

FIGURE 6: EXCHANGE RATE CLASSIFICATIONS (IRR 2019)

Panel A: Share of countries 1940-2015



Lecture plan:

- Fixed exchange rate (how?)
- Downside of fixed exchange rate (the impossible trinity)
- Different global monetary systems
- Global reserve currencies

Lecture plan:

- **Fixed exchange rate (how?)**
- Downside of fixed exchange rate (the impossible trinity)
- Different global monetary systems
- Global reserve currencies

To fix the exchange rate (three steps):

1. Set the target rate (or reset)
2. Change the value of the currency to the target rate
3. Keep the value of the currency at the target rate

<u>Supply of dollars</u>
Demand for pesos

<u>Demand for dollars</u>
Supply of pesos

$1= 12 pesos

INTERNATIONAL CURRENCY EXCHANGE

Supply of dollars
Demand for pesos

Demand for dollars
Supply of pesos

Devaluation

1$= 16 pesos

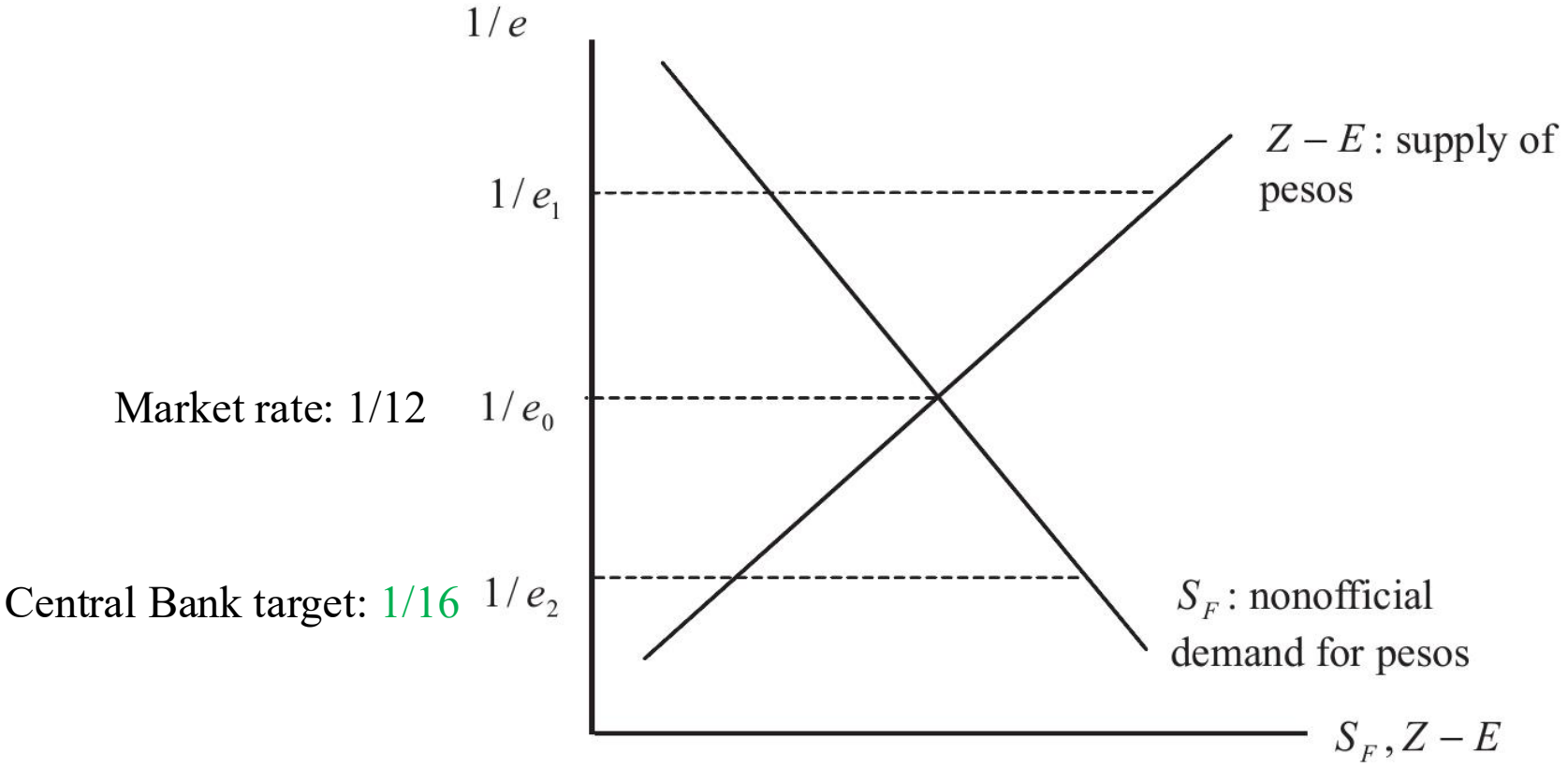


Figure 16.1. The Peso Market

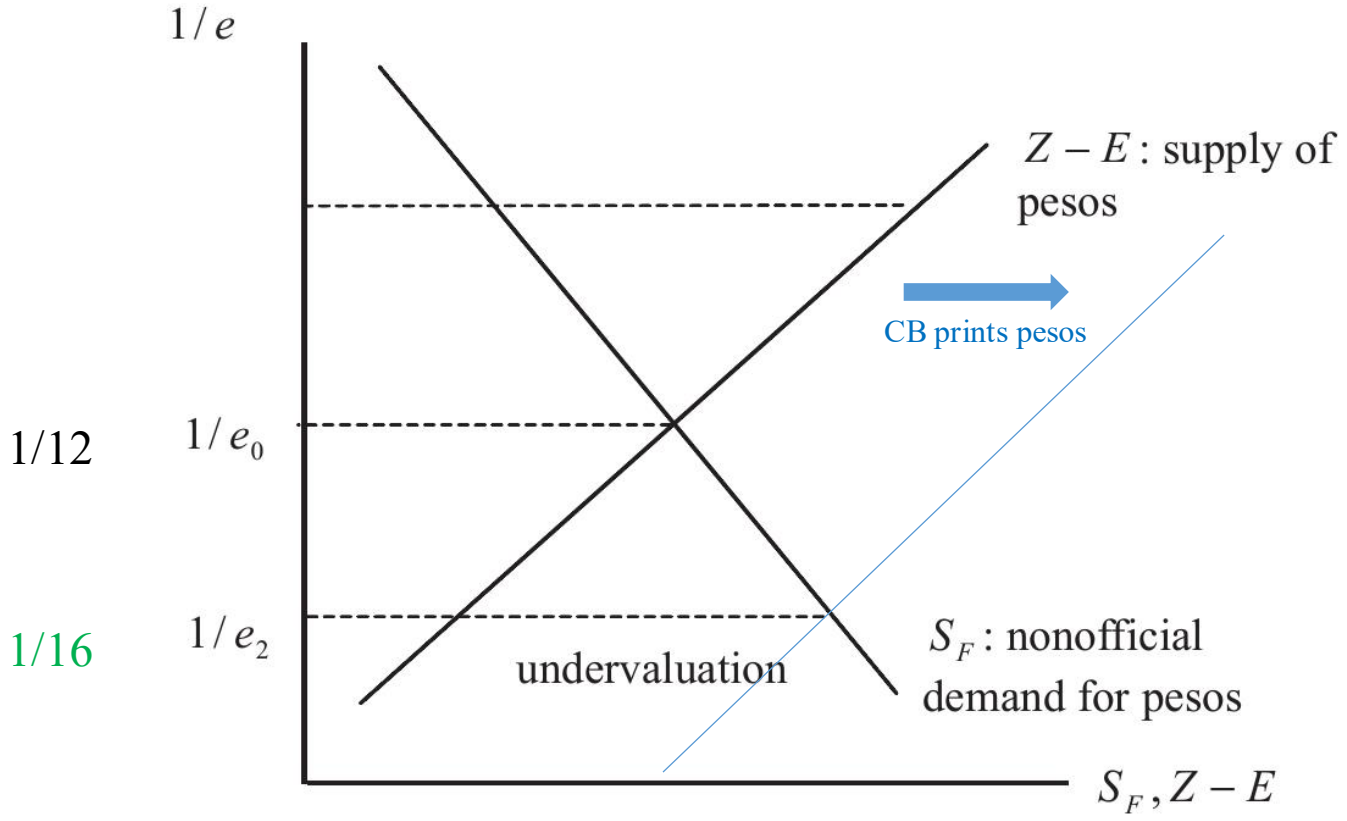


Figure 16.1. The Peso Market

<u>Supply of dollars</u>
Demand for pesos

<u>Demand for dollars</u>
Supply of pesos

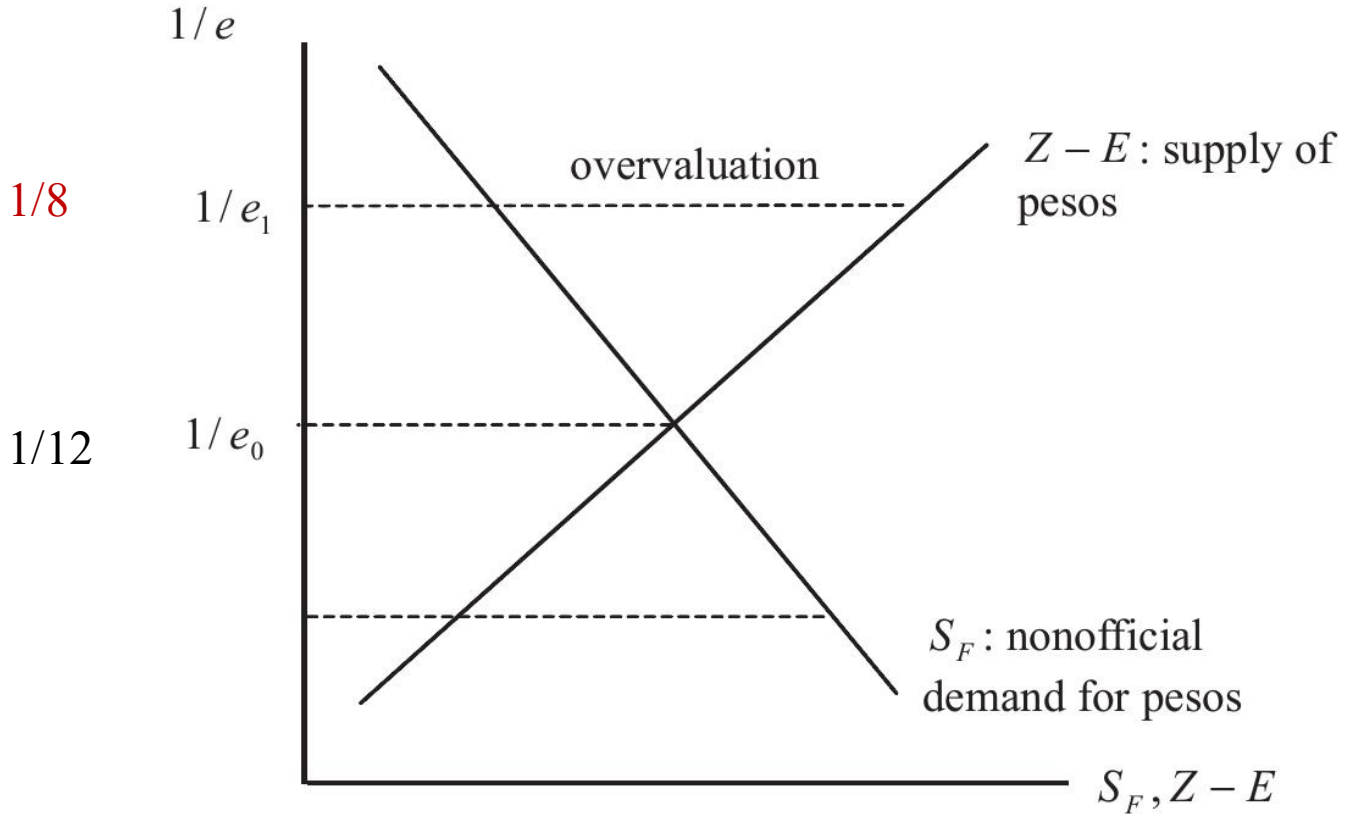
1$= 12 pesos

<u>Supply of dollars</u>
Demand for pesos

<u>Demand for dollars</u>
Supply of pesos

Revaluation

1$= 8 pesos



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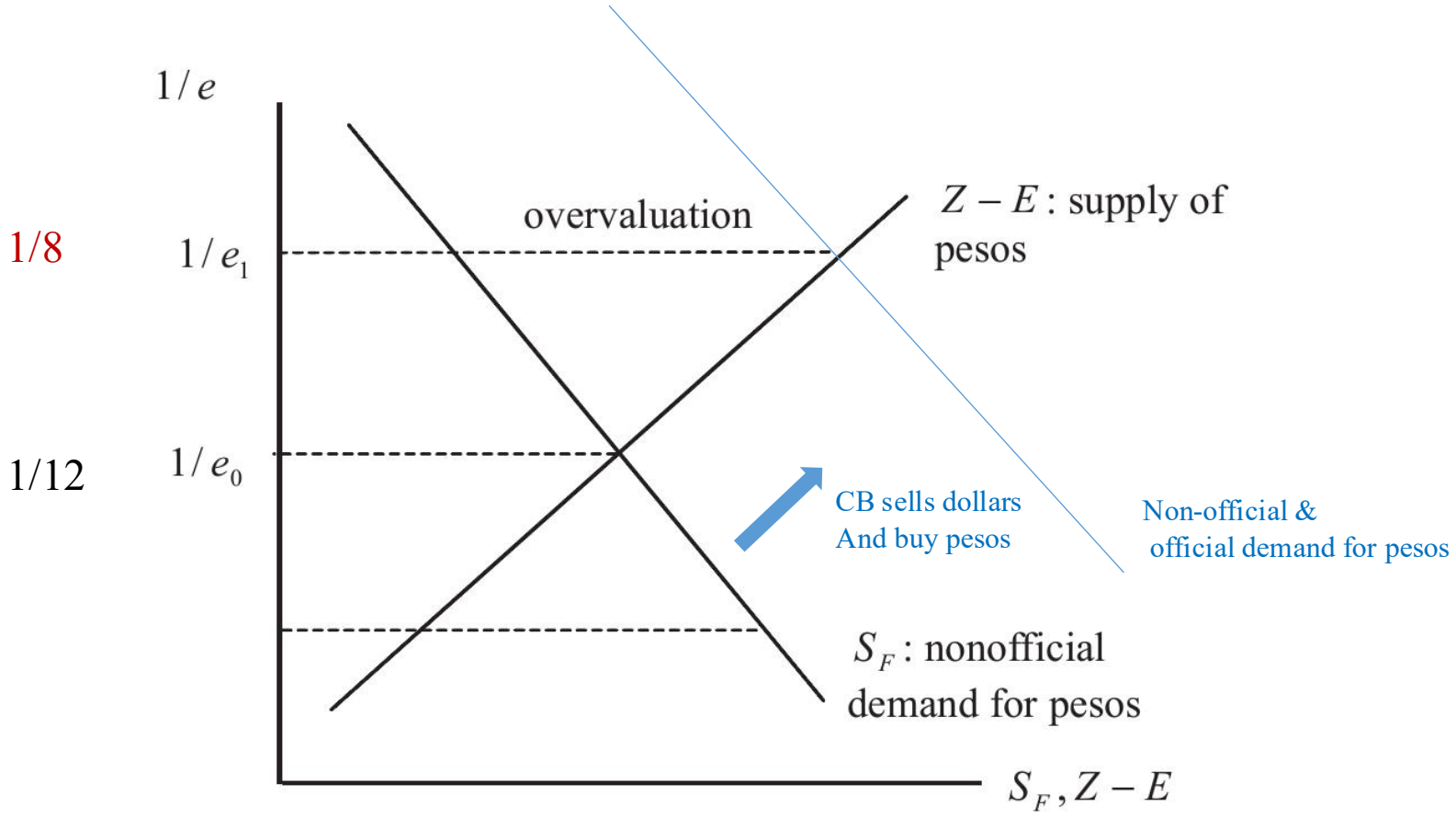

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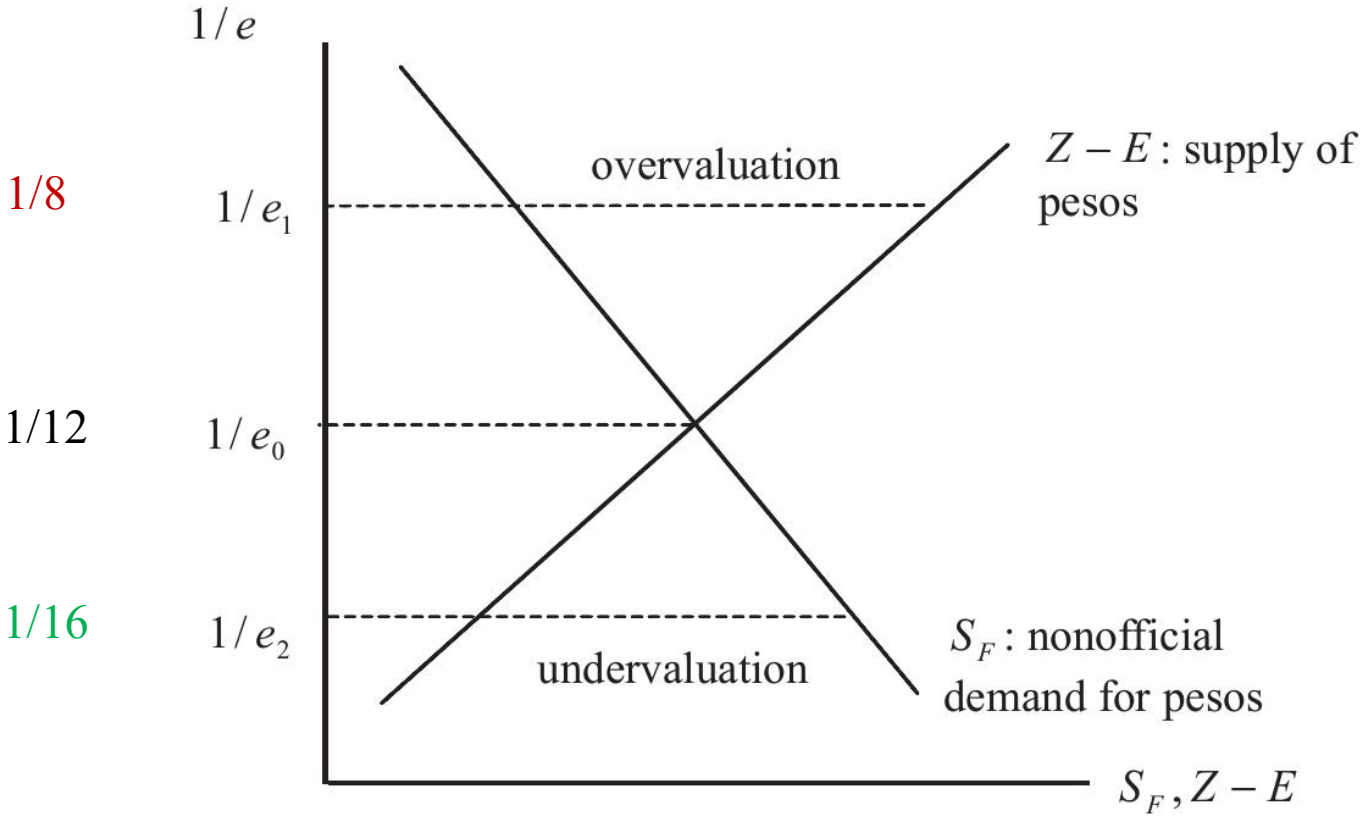
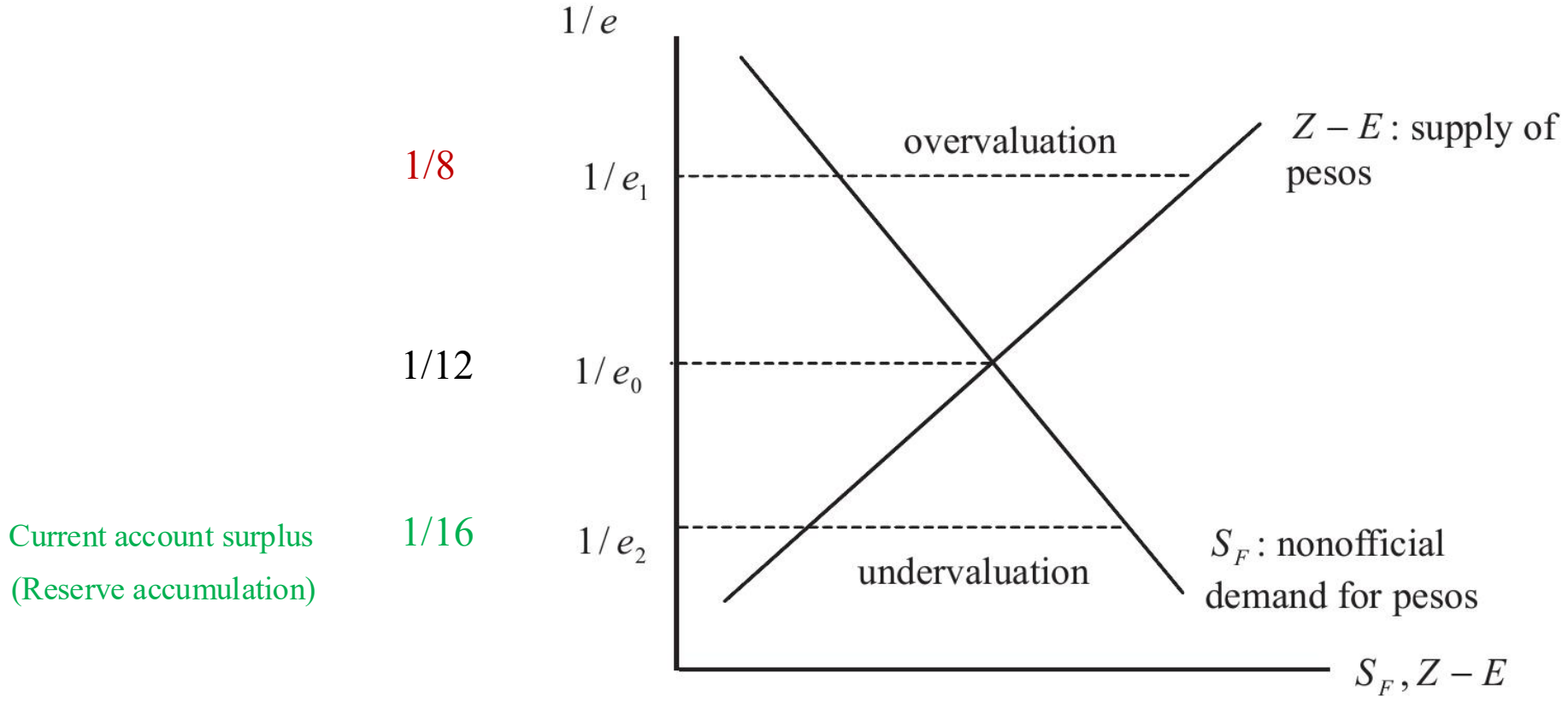


Figure 16.1. The Peso Market



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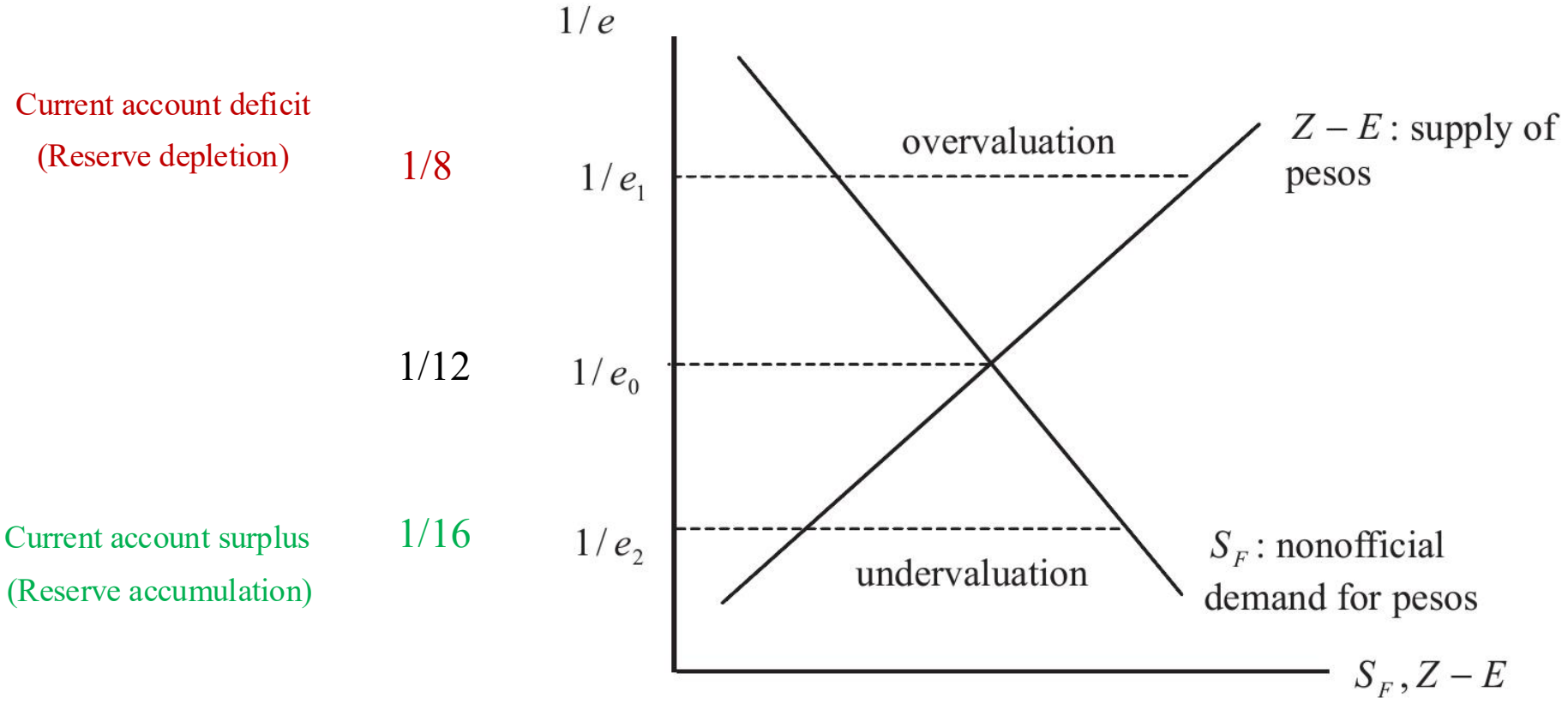
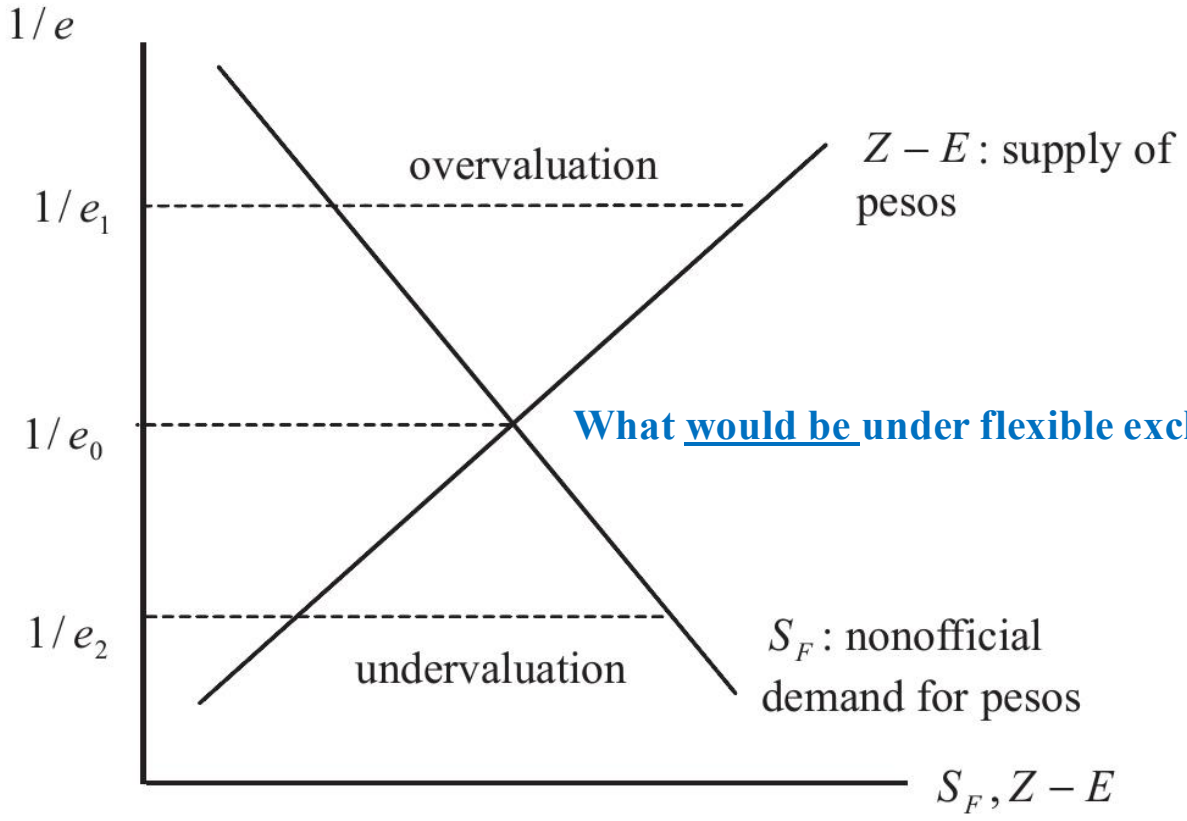


Figure 16.1. The Peso Market

Current account deficit (Reserve depletion) 1/8

Current account balance 1/12

Current account surplus (Reserve accumulation) 1/16



What would be under flexible exchange rate (PPP rate)

Figure 16.1. The Peso Market

Lecture plan:

- Fixed exchange rate (how?)
- **Downside of fixed exchange rate (the impossible trinity)**
- Different global monetary systems
- Global reserve currencies

Black Wednesday (BBC 1997)

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Lecture plan:

- Fixed exchange rate (how?)
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- Different global monetary systems
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Trade-based model of fixed exchnage rates:

Inflation differences across countries → Trade flows $\xrightarrow{?}$

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Inflation differences across countries → Trade flows $\xrightarrow{?}$

Inflation is 8% higher in Mexico

$$\frac{\text{Domestic prices in US (\$)}}{\text{Domestic prices in Mexico (pesos)}} \times \overset{\textit{Fixed}}{e} = \frac{\text{Domestic prices in US (Pesos)}}{\text{Domestic prices in Mexico (Pesos)}}$$

8% inflation

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Trade deficit in Mexico

Trade-based model of fixed exchange rates:

Inflation differences across countries → Trade flows → No <u>automatic</u> adjustment

Inflation is 8% higher in Mexico

Trade deficit in Mexico

Trade-based model of fixed exchnage rates:

Inflation differences across countries → Trade flows → No automatic adjustment

Asset-based model of fixed exchange rates:

Interest rates differences across countries → Capital flows

Interest rate is 3% higher in the US

Trade-based model of fixed exchnage rates:

Inflation differences across countries → Trade flows → No automatic adjustment

Asset-based model of fixed exchange rates:

Interest rates differences across countries → Capital flows

Interest rate is 3% higher in the US

Capital account deficit in Mexico

Trade-based model of fixed exchnage rates:

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Trade-based model of fixed exchange rates:

Inflation differences across countries → Trade flows → No automatic adjustment

Inflation is 3% higher in Mexico

Trade deficit in Mexico

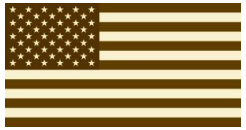
Asset-based model of fixed exchange rates:

Interest rates differences across countries → Capital flows → No automatic adjustment

Interest rate is 3% higher in the US

Capital account deficit in Mexico

Question: Now, what should Mexican policymakers do to ensure their capital account (and current account) get in balance?



2%

$5 billion
(60 billion pesos)

Investor

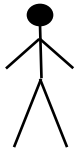
Fixed
1$= 12 pesos

INTERNATIONAL CURRENCY EXCHANGE

Target:
$= 12 pesos

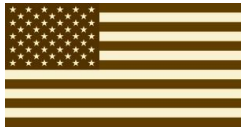
CB

2%



Investor

$5 billion
(60 billion pesos)



Now interest rate in the US is 3% higher.

5%

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INTERNATIONAL CURRENCY EXCHANGE

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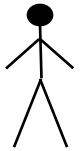

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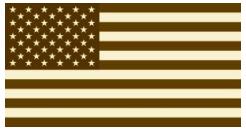


Now interest rate in the US is 3% higher.

5%

$7 billion
(84 billion pesos)

Investor

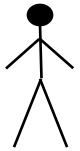
Fixed
1$= 12 pesos

INTERNATIONAL CURRENCY EXCHANGE

Target:
$= 12 pesos

CB

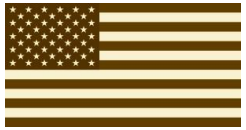
2%



Investor

$5 billion
(60 billion pesos)

Option one: Mexican policymakers increase their interest rates to the level of the US.



5%

$7 billion
(84 billion pesos)

Investor

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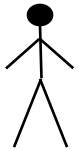
1$= 12 pesos

INTERNATIONAL CURRENCY EXCHANGE

Target:
$= 12 pesos

CB

5%



Investor

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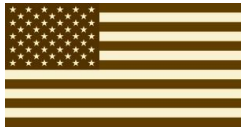

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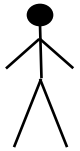


Investor

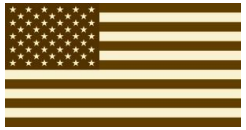
$5 billion
(60 billion pesos)

Option one: Mexican policymakers increase their interest rates to the level of the US.

But you end up sacrificing your monetary policy independence

There is another alternative tho!

Option two: Mexican policymakers can introduce restriction on capital outflows rather than raising their interest rates



5%

Restriction

$7 billion
(84 billion pesos)

Investor

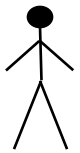
Fixed
1$= 12 pesos

ice INTERNATIONAL CURRENCY EXCHANGE

Target:
$= 12 pesos

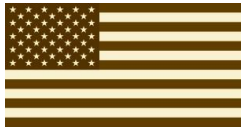
CB

2%



Investor

$5 billion
(60 billion pesos)



5%

Restriction

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(84 billion pesos)

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Target:
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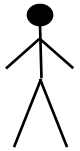
CB

2%



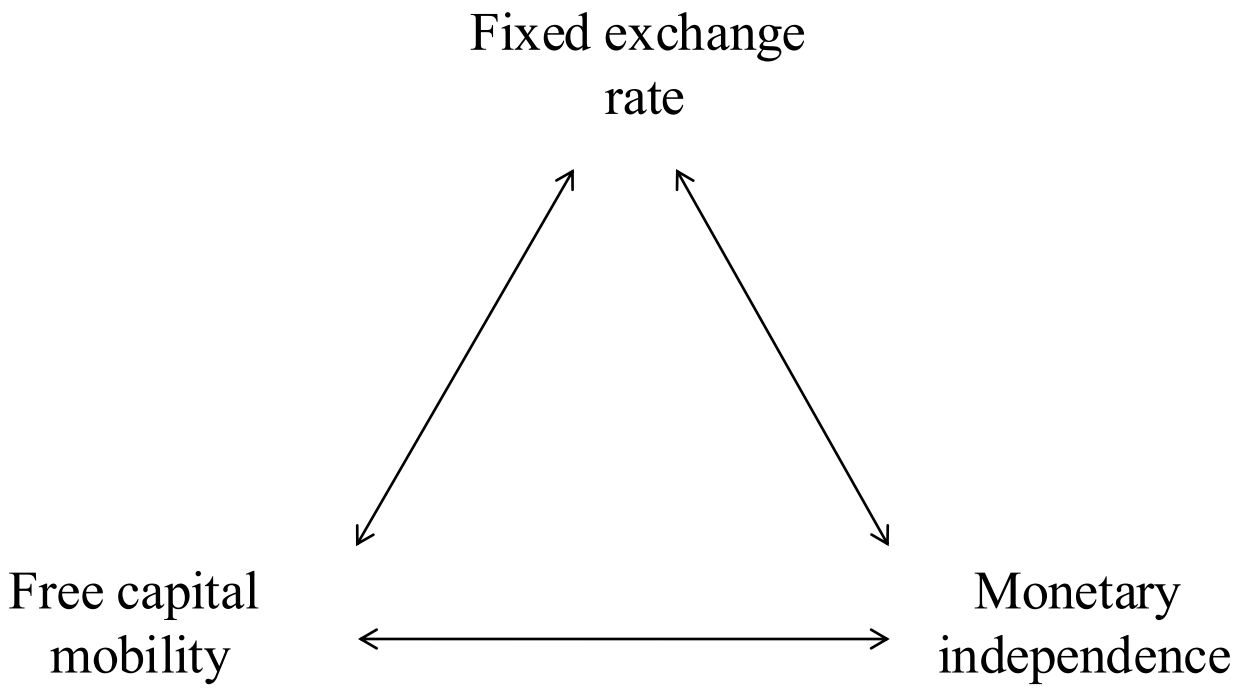
Investor

$5 billion
(60 billion pesos)

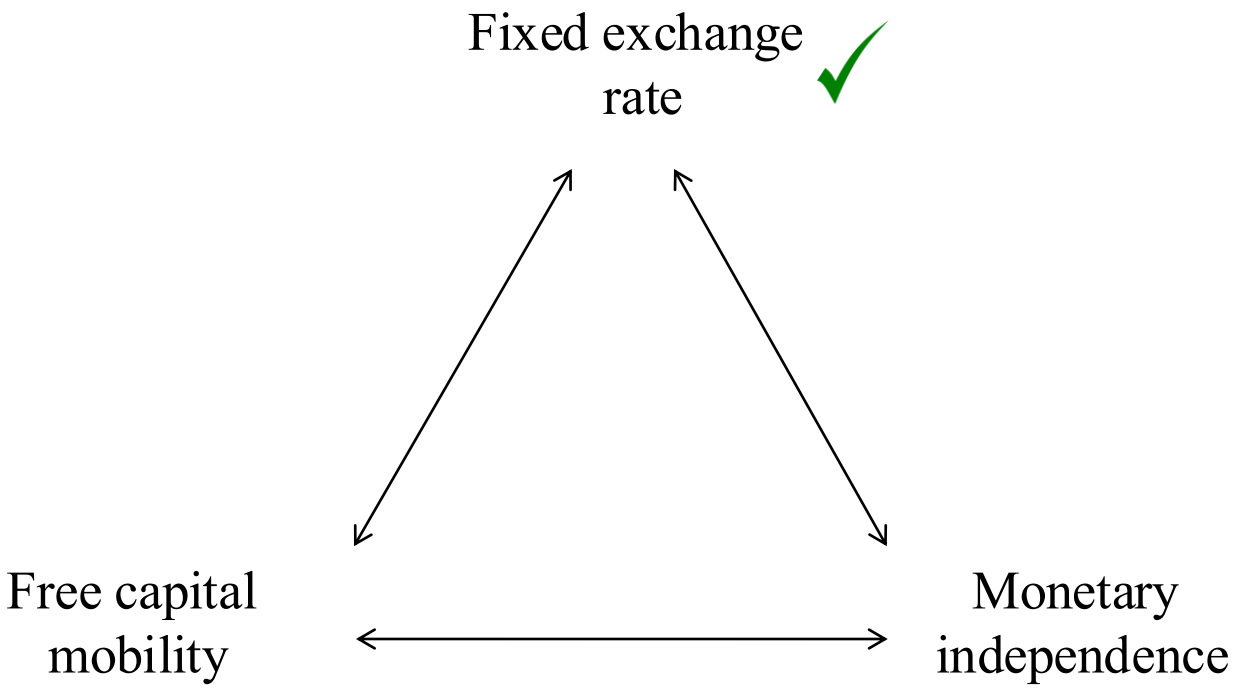
Option two: Mexican policymakers can introduce restriction on capital outflows

But they end up sacrificing their openness to inflows and outflows of capital

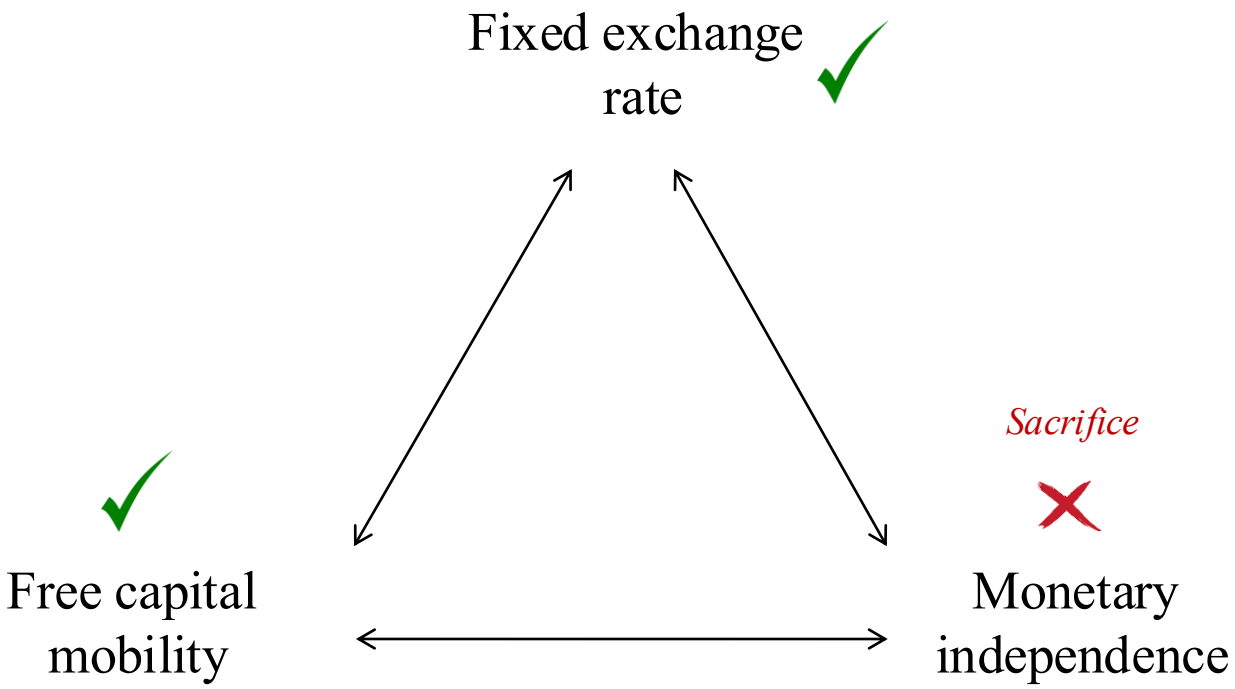
Impossible trinity



Impossible trinity



Downside of having a fixed exchnage rate: you have to sacrifice your monetary independence *or* your openness to inflows and outflows of finance.


Fixed exchange rate
Free capital mobility
Sacrifice
Monetary independence

Fixed exchange rate ✓

Sacrifice

✗

Free capital mobility

✓

Monetary independence

Fixed exchange rate

Sacrifice

✓ Free capital mobility

✓ Monetary independence

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Saudi Central Banker Says Crunch That Squeezed Banks Is Over

→ Saudi central bank chief spoke in an interview in Davos

Alkholifey on Opening Up Saudi Markets, Economy

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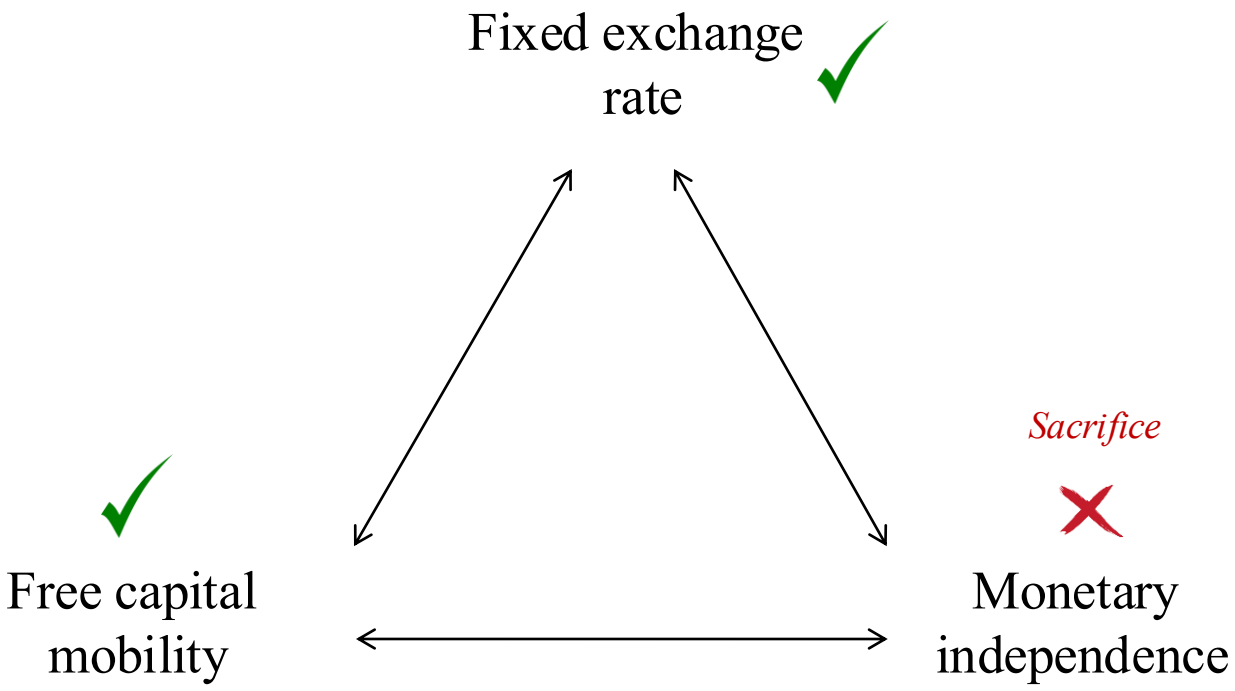
Lecture plan:

- Fixed exchange rate (how?)
- Downside of fixed exchange rate (the impossible trinity)
- **Different global monetary systems**
- Global reserve currencies

Gold Standard (1870-1914)



What was the adjustment mechanism under the Gold Standard? What would a country do if they experienced a current account deficit due to a drop in external demand for their goods?

Professor Keynes is Optimistic

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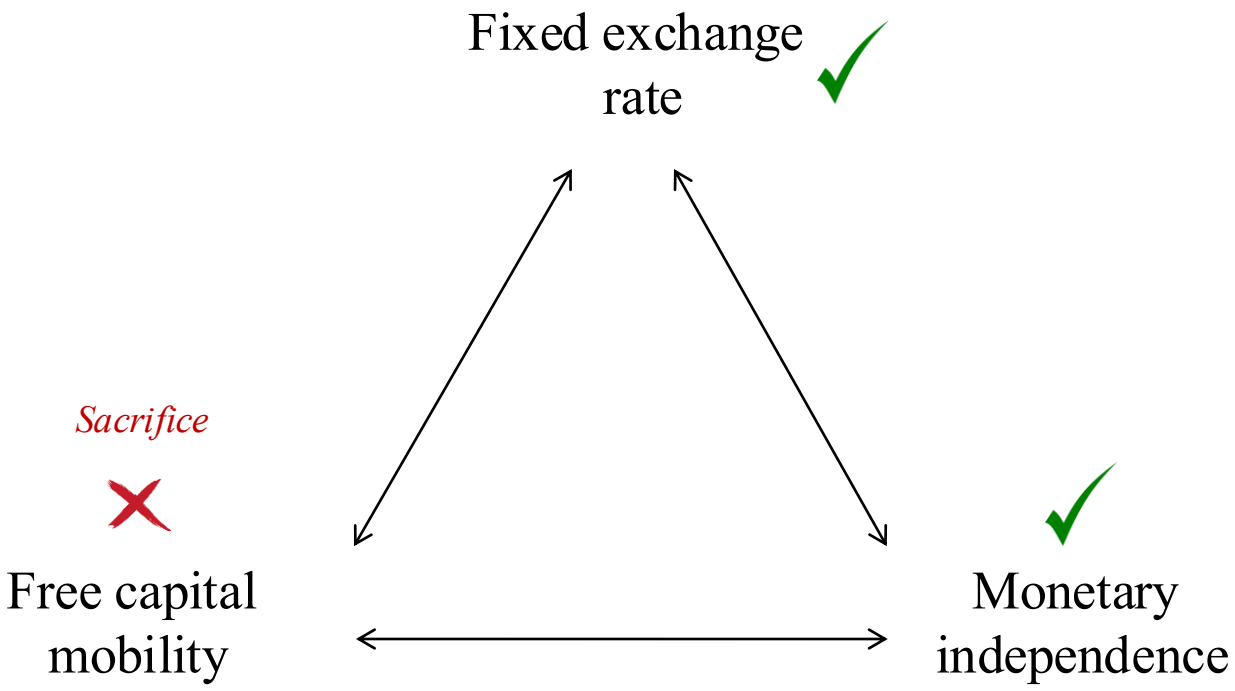
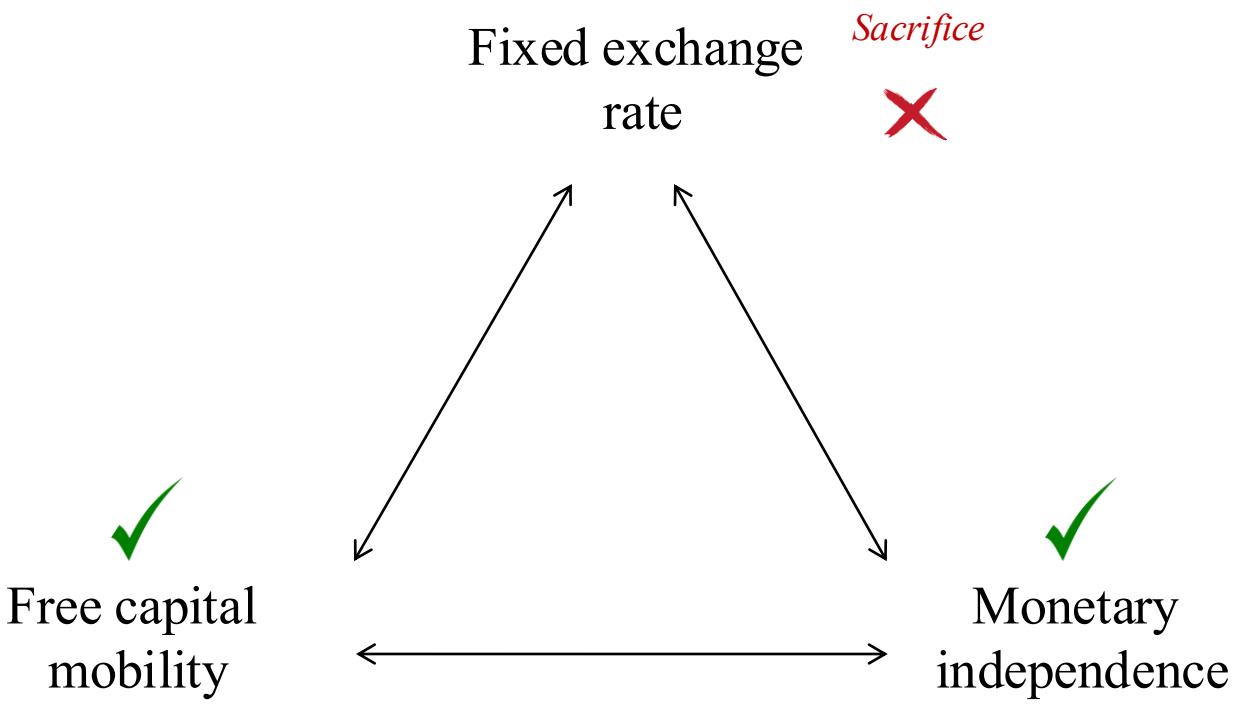
Bretton Woods (1945-1971)



Post Bretton Woods (1971-now)



What was the adjustment mechanism under the Bretton Woods regime? What would a country do if they experienced a current account deficit?

Lecture plan:

- Fixed exchange rate (how?)
- Downside of fixed exchange rate (the impossible trinity)
- Different global monetary systems
- **Global reserve currencies**

Global reserve currency

- Unit of account: Many global market, oil for example, are priced in dollars.

- Means of payment: Let’s say India and Brazil do their trade with each other in dollars.

- Store of value: Countries keep their savings in dollar.

FIGURE 6: EXCHANGE RATE CLASSIFICATIONS (IRR 2019)

Panel A: Share of countries 1940-2015

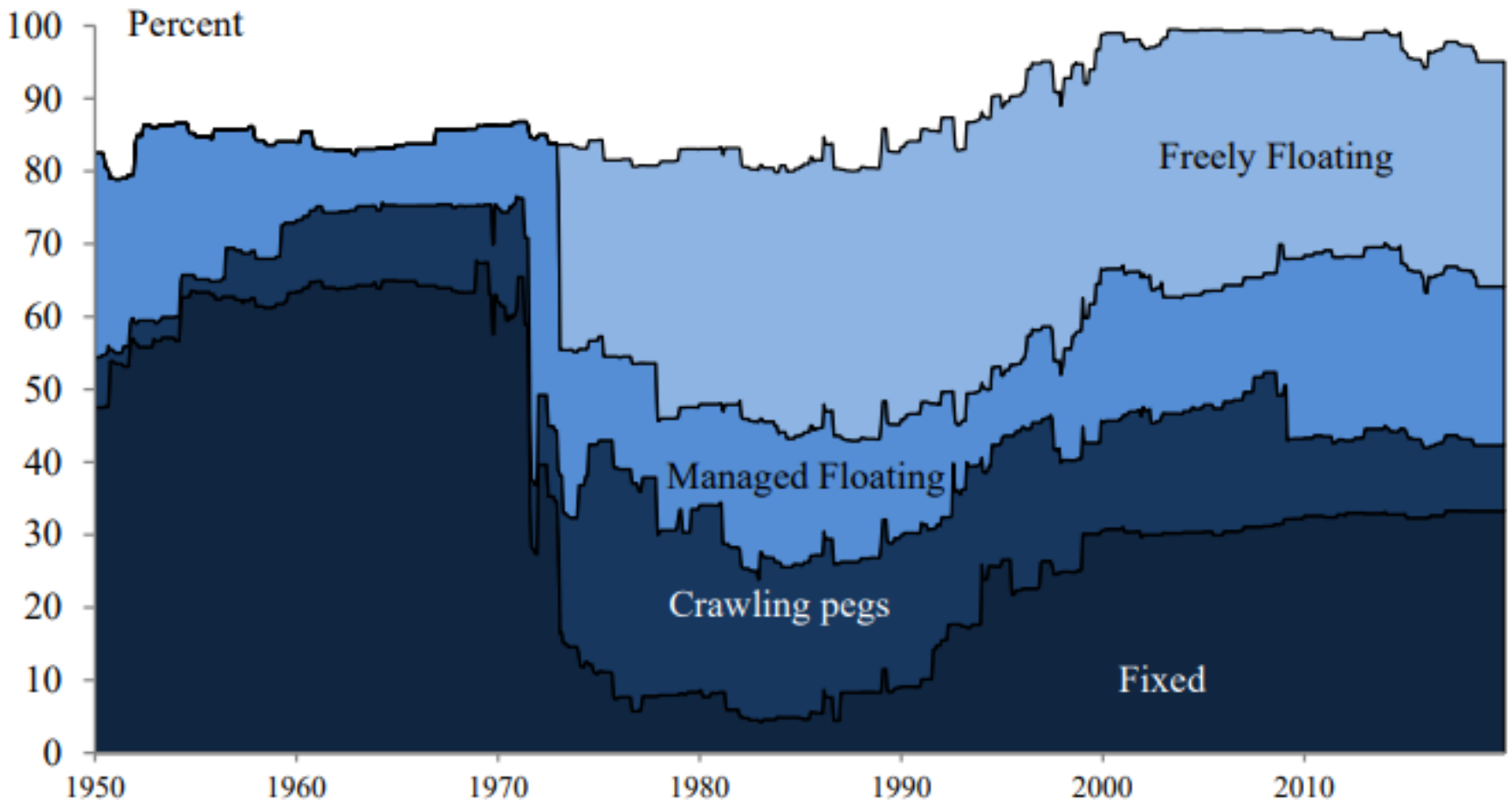


FIGURE 2: ANCHOR CURRENCY CLASSIFICATIONS (IRR, 2019, 2021 UPDATE)

Panel A: Share of countries 1946-2019

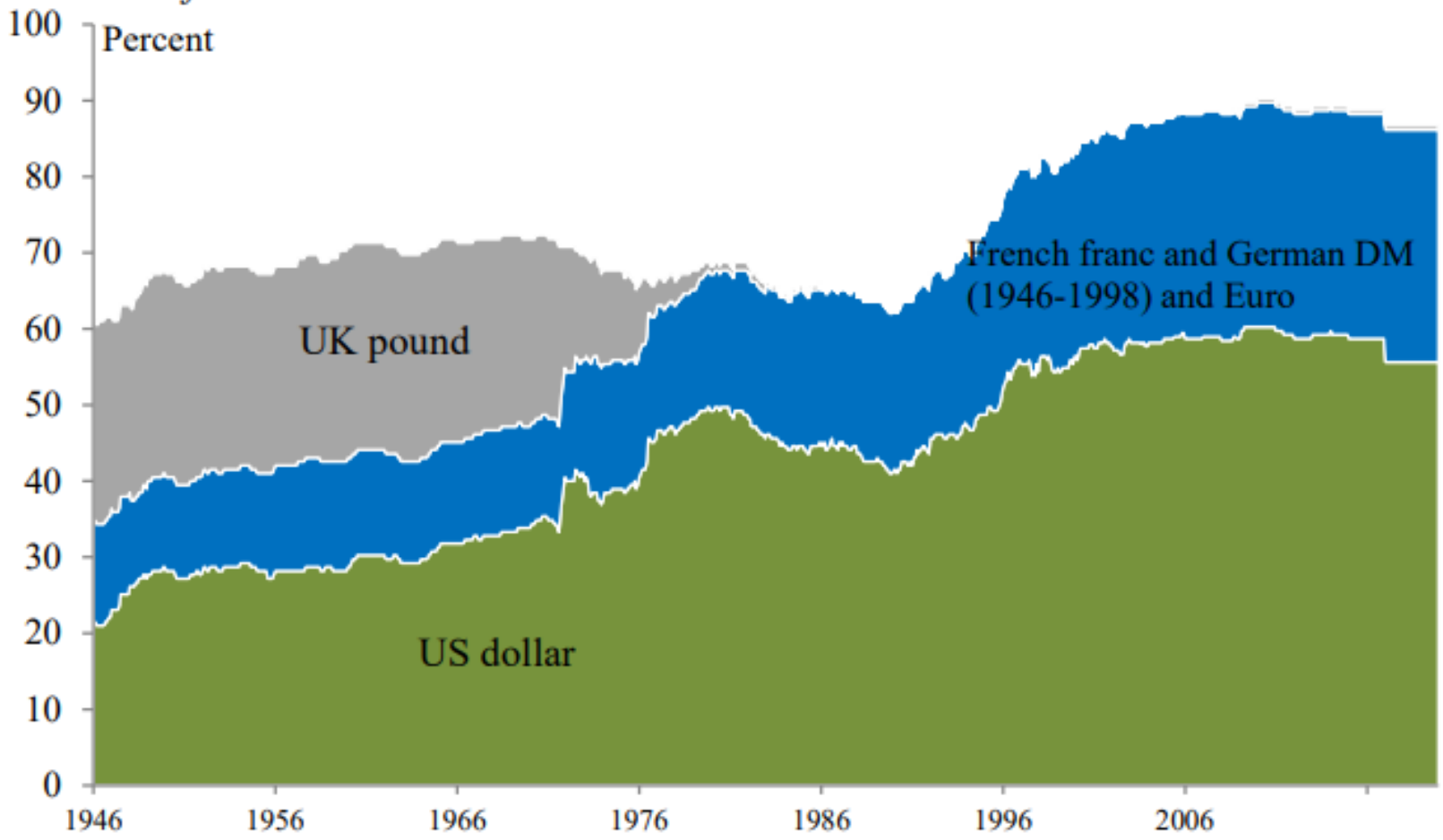
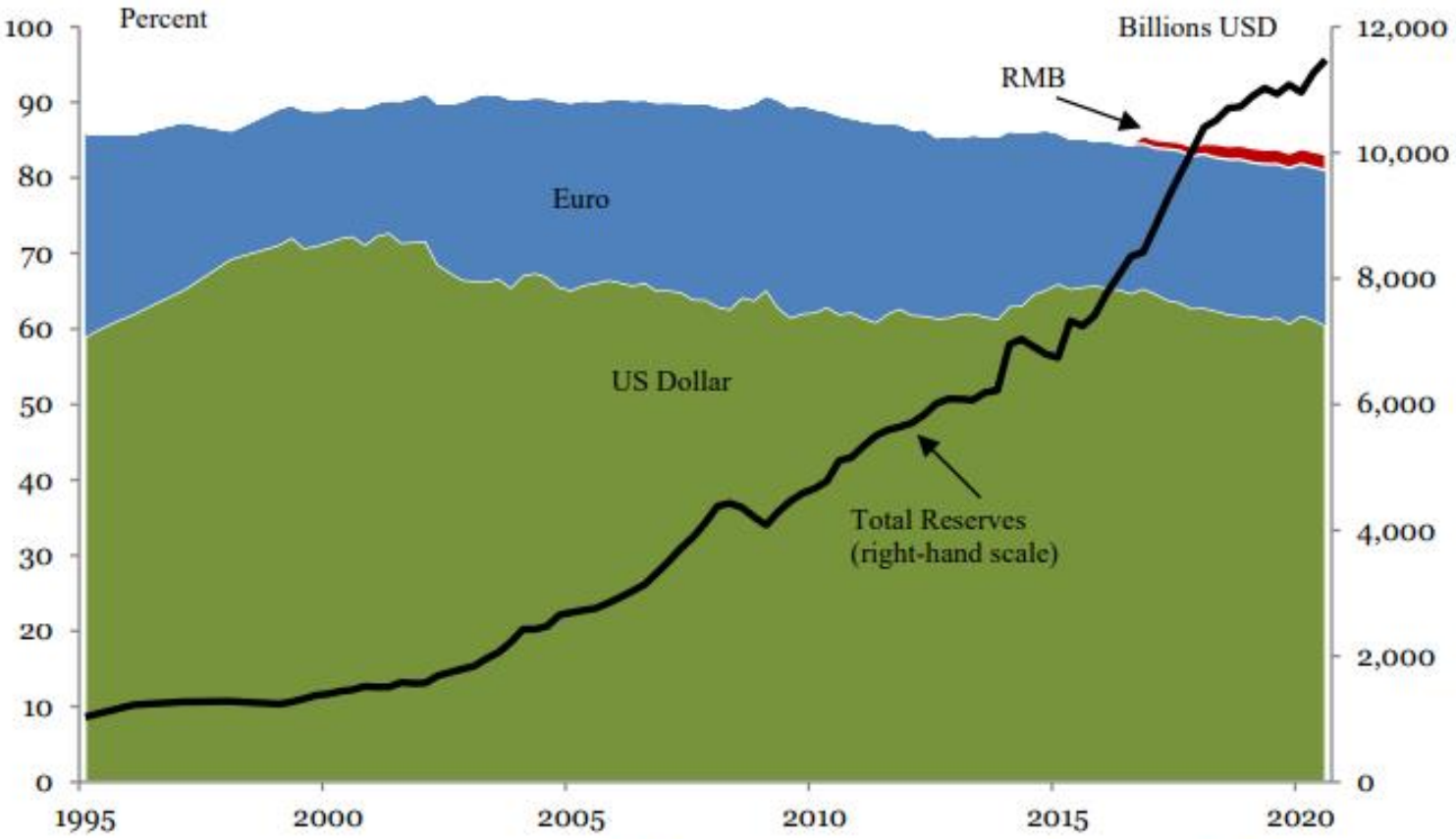


FIGURE 7: CENTRAL BANK FOREIGN CURRENCY RESERVES



Note: The solid line shows the total stock of world central bank reserves in billions of US dollars (right-hand scale). The shaded areas show the share of central bank reserves by currency denomination in percent of total (left-hand scale). Sources: Ilzetzki, Reinhart and Rogoff (2020a), IMF Cofer and the authors.

	USA	China
Total GDP (PPP)	$30 Trillion	$40 Trillion
GDP per capita (PPP)	$90,000	$30,000
Share of world GDP	15%	19%
Share of world trade	10%	14%
Share of reserve currency	60%	5%

Source: IMF Data

What are the advantages of having a global reserve currency?

U.S. Debt
% of GDP
300
250
200
150
100
50
0
The U.S. financial crisis began in 2007 with total debt close to 230% of GDP
S&L Crisis
Non-Financial Corporations
Household
Government
1952
1956
1960
1964
1968
1973
1977
1981
1985
1989
1993
1998
2002
2006
2010
2014
2018
2023

U.S. Balance of Payments 1980 to 2010

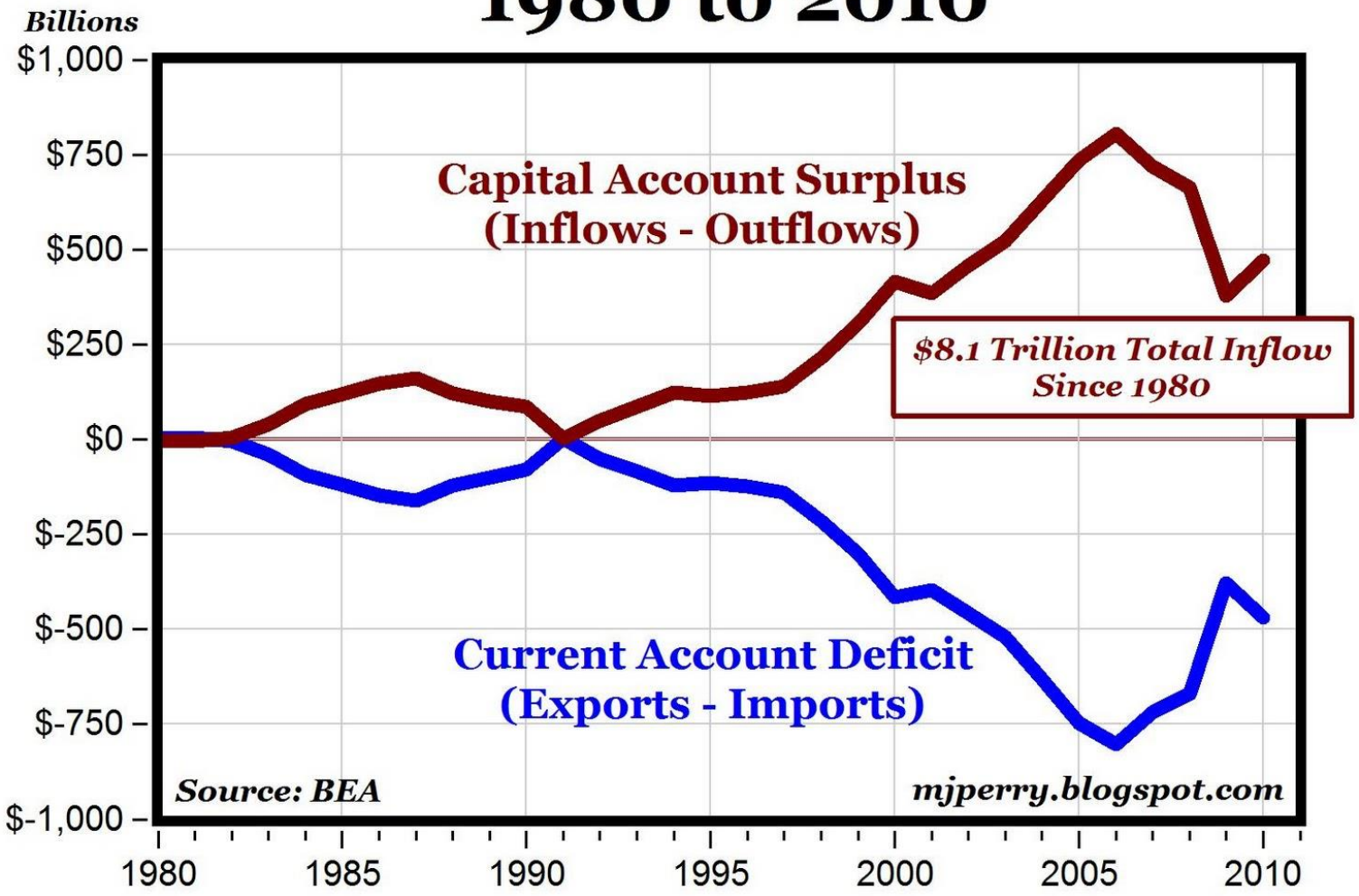
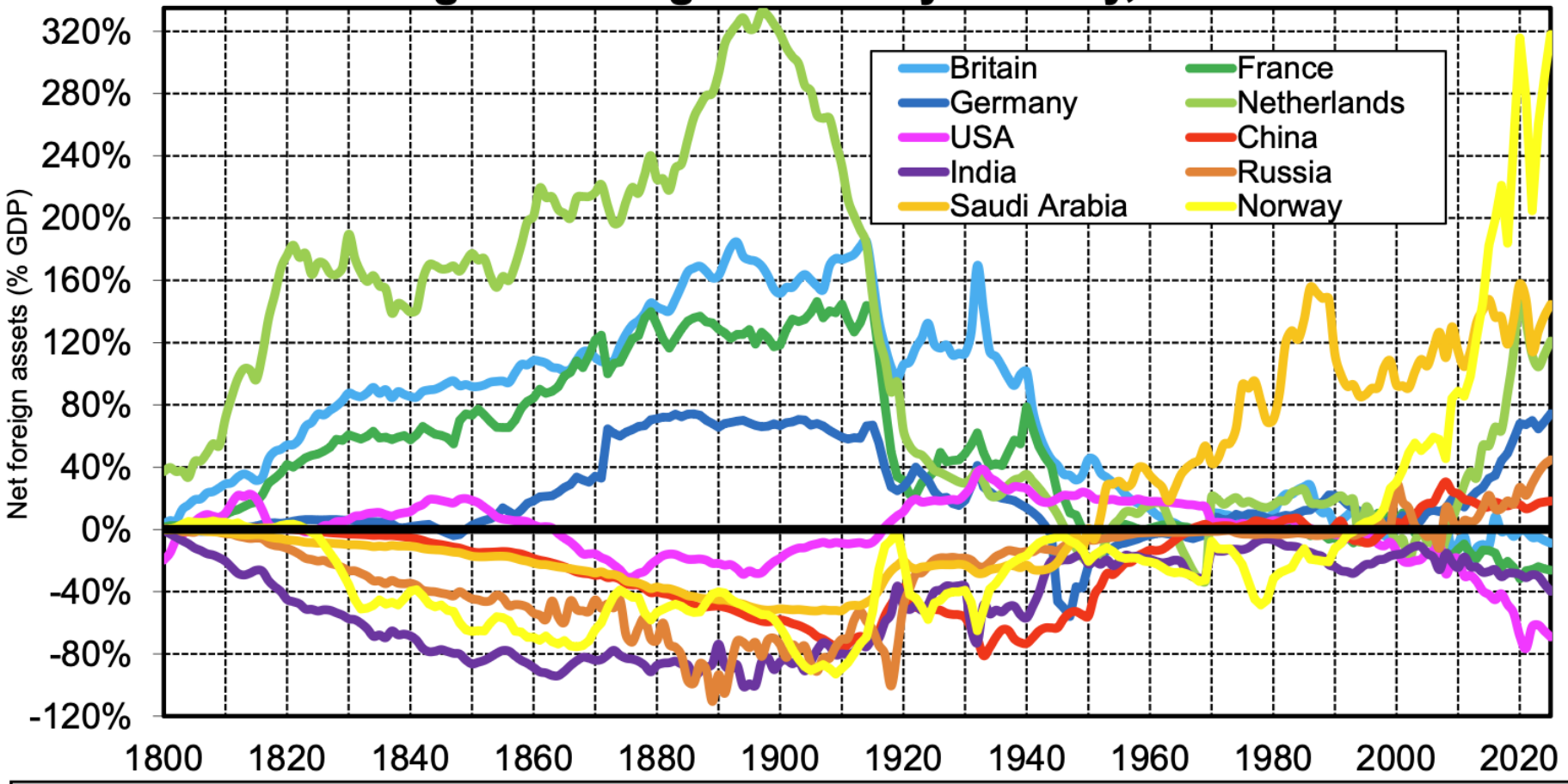
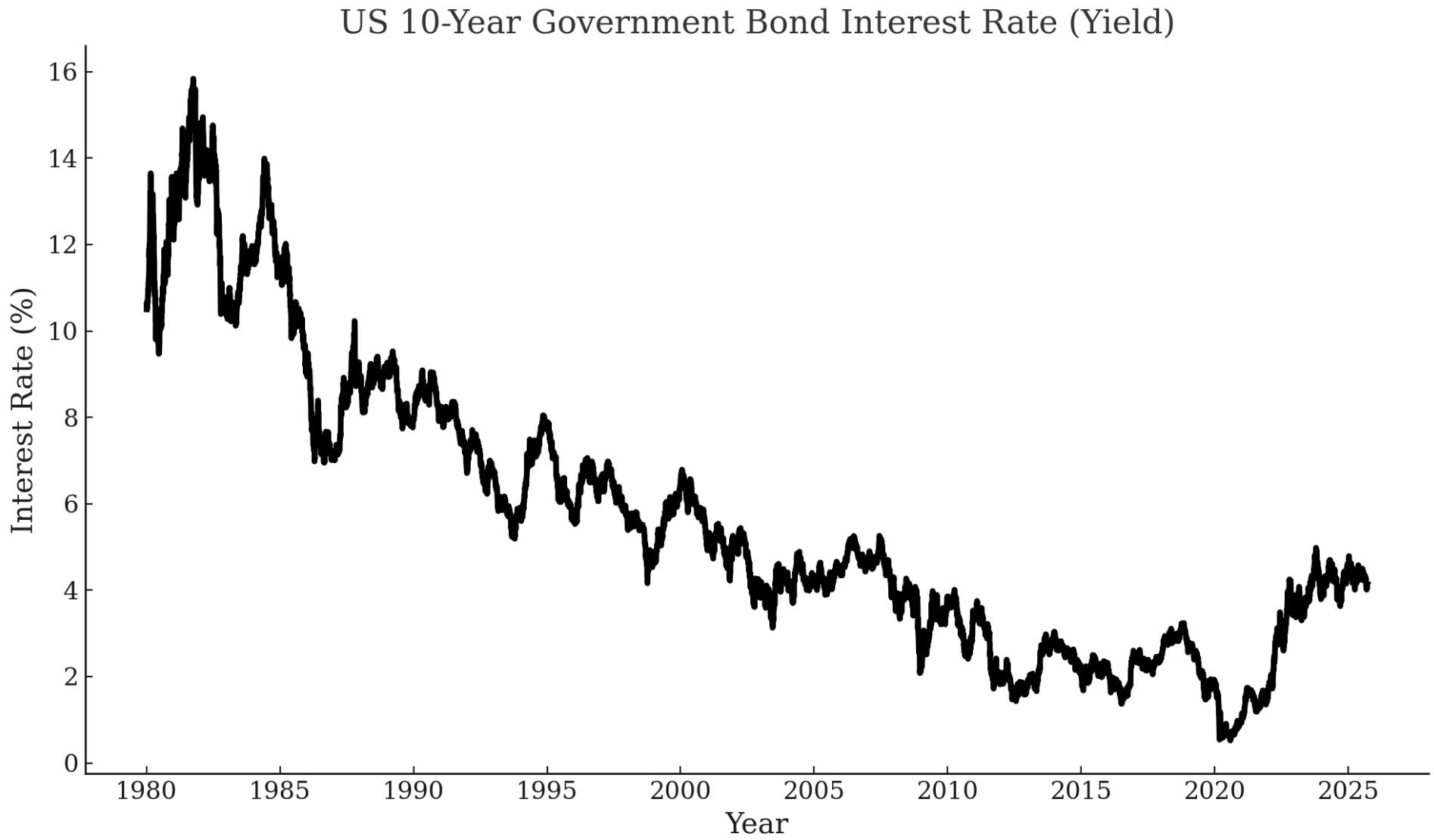
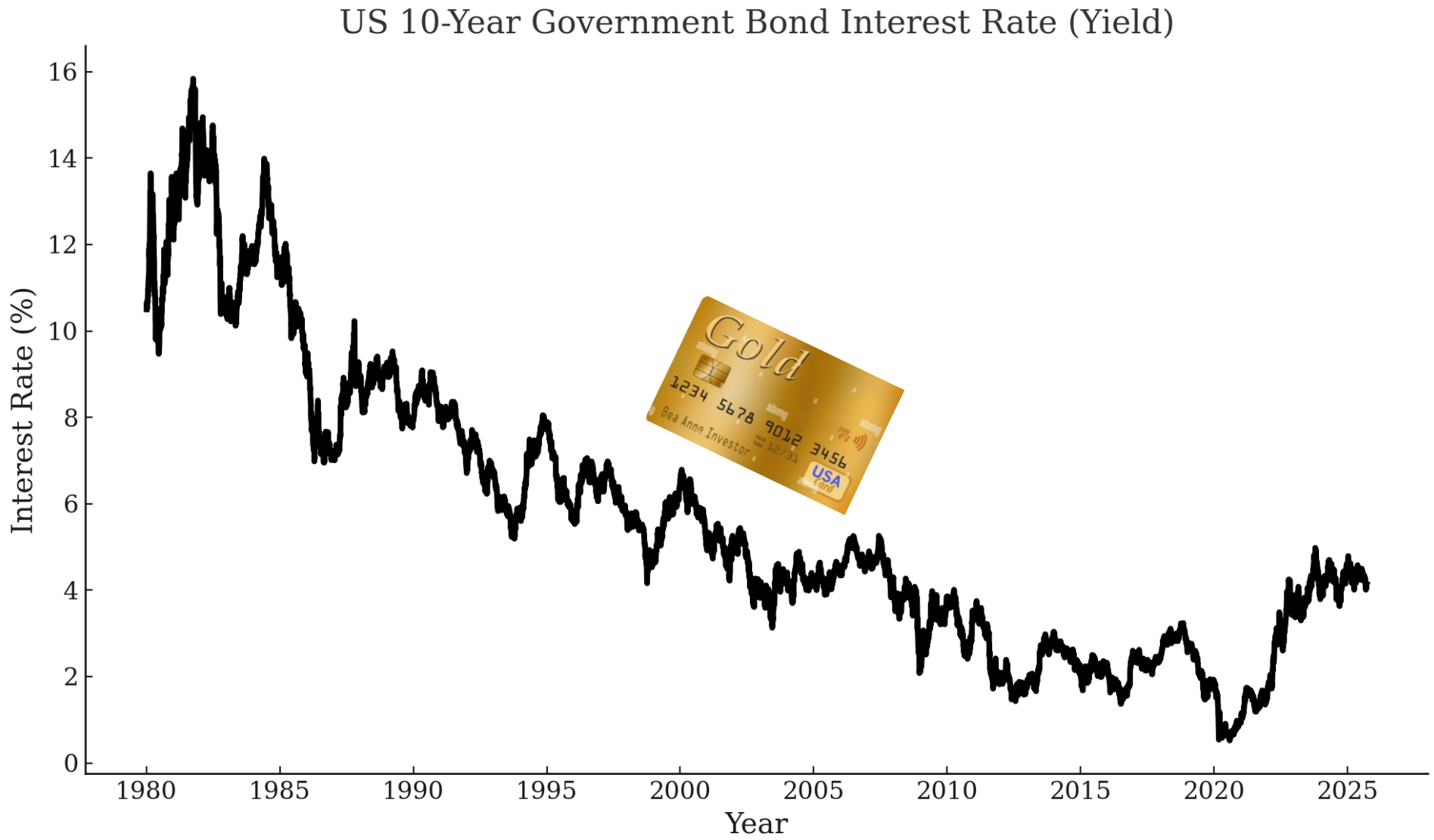


Fig. 14. Foreign Wealth by Country, 1800-2025



Interpretation. If we include smaller economies into the picture, we find that net foreign assets can be as large as 300% of a country's GDP or more, such as the Netherlands in 1900 (a small country with large colonial holdings in Indonesia) or Norway in 2025 (a small country with enormous oil and gas reserves that were transformed into a large sovereign fund in a recent decades).
Sources and series: wid.world


US 10-Year Government Bond Interest Rate (Yield)
Interest Rate (%)
0
2
4
6
8
10
12
14
16
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year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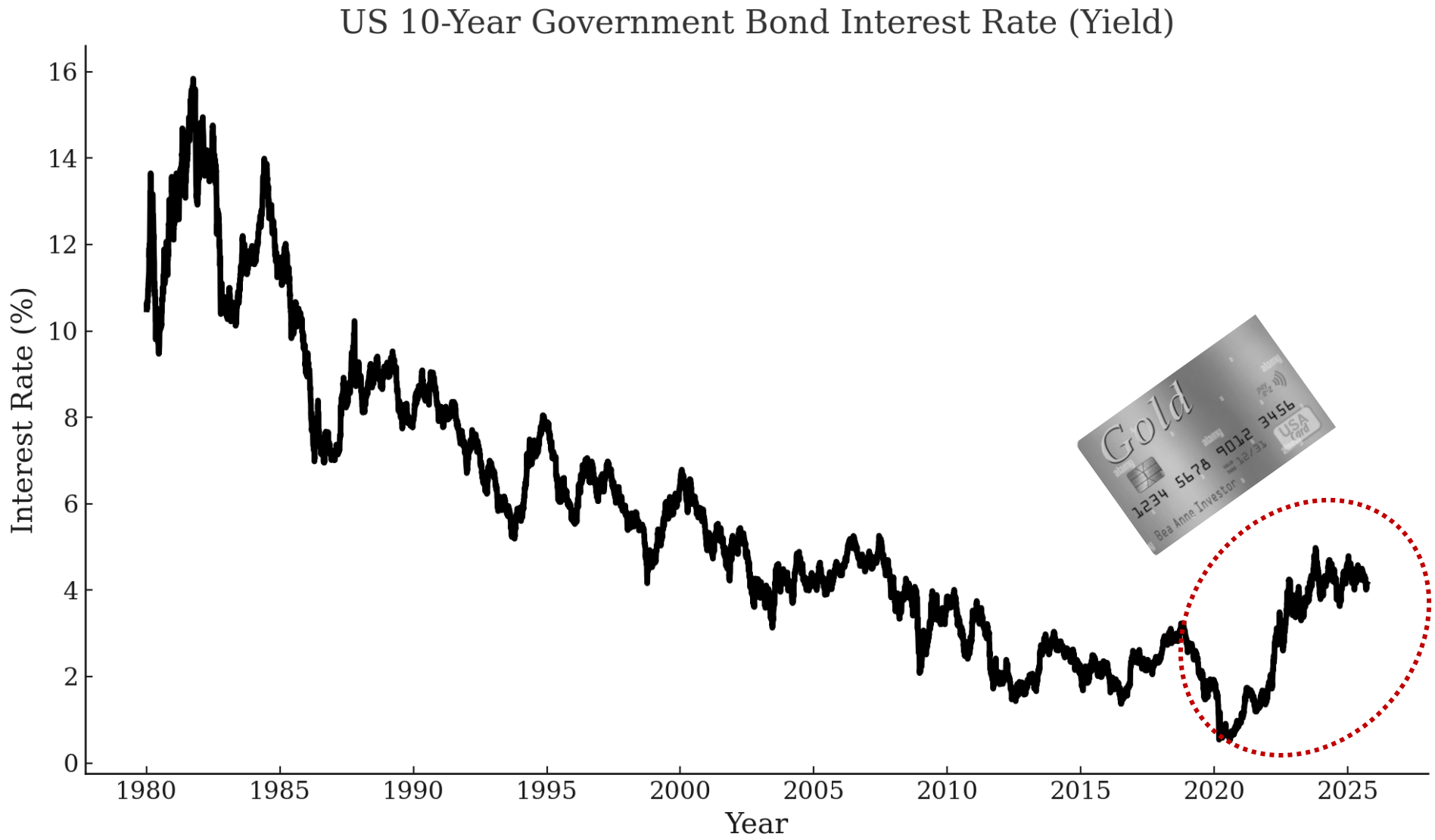
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- To have a fixed exchange rate, countries have to sacrifice either their monetary independence or openness to inflows and outflows of investment (impossible trinity) to ensure balance in their capital & current accounts.

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- Impossible Trinity explains different global monetary arrangements throughout history.